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Basic LaTeX formatting

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Now you know how to write a report, it is time to learn how you can do this using the ***L^AT_EX***.

We will start with discussing the overall structure of a ***L^AT_EX*** document, for which we will use this template: Template_Latex_Report.tex

You need a photo as well for this template: Rainbowfish1000x774.jpg. Download this photo and put it into the same folder as the tex file. That way, Latex will know where to find it.

When you compile this template, the result should be: Template_Latex_Report.pdf.

Note: Always compile your ***L^AT_EX*** document three times using PDF-latex, to ensure that all internal references are processed correctly.

In the template you might notice the symbol % plays a special role: It is the ***L^AT_EX*** symbol to indicate comments. So everything on a line after a % will not be displayed by ***L^AT_EX*** in the compiled document.

Structure of a Latex document

The overall structure of every ***L^AT_EX*** document is the same, and must look like this:

```
\documentclass{...}
...
\begin{document}
...
\end{document}
```

The *preamble* are the lines between `\documentclass{...}` and `\begin{document}`.

Between `\begin{document}` and `\end{document}`, you put the content (i.e. text) of your

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The preamble

In the preamble you define the following things:

- The type of document template you will use;
- The packages you will use in your document. Packages give you the option to use more commands in the document;
- The (extra) layout options you want;
- All other stuff that is not the content of your report.

Exercise A

1/1 point (graded)

In the preamble a package is loaded that will help you in using graphs.

On which line in the template is this package loaded?

✓ Answer: 11

Submit

You have used 1 of 10 attempts

i Answers are displayed within the problem

The title page

In the template, we have chosen to define our own title page. This title page is defined between `\begin{titlepage}` on line 22 and `\end{titlepage}` on line 55. Between these two lines, you can change several things to make your own title page.

On line 29, the photo is imported by ***L^AT_EX***. If you want to use another photo, put it in the same folder as the template and change the filename of our photo to the filename of your photo.

On lines 32 and 33 you can change the title and subtitle of the report.

Exercise B

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Which four lines in the template do you need to change for this? Give the line numbers in increasing order.

✓ Answer: 40

✓ Answer: 41

✓ Answer: 48

✓ Answer: 49

You have used 1 of 10 attempts

i Answers are displayed within the problem

Adding chapters

Now you have a title page, it is time to add chapters to your report.

You can add two types of chapters in *L^AT_EX*: numbered and unnumbered.

Numbered chapters you obtain by using the command `\chapter{Title of chapter}`. This command will start a new page, typeset the title of the chapter and add the chapter to the table of contents.

Unnumbered chapters you obtain by using the command `\chapter*{Title of chapter}`. This command will start a new page, typeset the title of the chapter but will not add the chapter to the table of contents. So if you want this chapter in the table of contents, you must add it manually: You add the line `\addcontentsline{toc}{chapter}{Title of chapter}`. You can put this line right behind the opening command of your chapter. See for

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Exercise C

1/1 point (graded)

Speaking of table of contents: you want that in your report as well. In the template, there is one line that commands Latex to print the table of contents.

What command prints the table of contents?

Do not include the comments behind the command.

✓ Answer: \tableofcontents

Submit

You have used 1 of 10 attempts

📘 Answers are displayed within the problem

More structure for your document

After you have made your chapters, you might want to add some sections, and perhaps even subsections, in these chapters. In ***L^AT_EX*** you can use the commands `\section{Title of section}` and `\subsection{Title of subsection}` for this. If you want unnumbered sections, you can add a `*` between the `n` and the `{}`.

You might want to refer back or forward to a chapter or section in your document. To make ***L^AT_EX*** do this, there are two command you need to use. First, put the command `\label{tagname here}` behind the command that defines the chapter/section. Then you can put `\ref{tagname here}` at exactly the place in the text you want the reference to the chapter/section to appear. This last command only prints the number of the chapter/section, so you should add "Chapter" or something similar yourself.

Exercise D

1/1 point (graded)

Another way to structure your document and text is to use lists. ***L^AT_EX*** provides you with two standard options, just like chapters and section: numbered and unnumbered lists. You can create an item in such a list by the command `\item`. You must add such items between a `\begin{...}` and a `\end{...}` to get (un)numbered lists, where `...` is replaced with the name of the list in ***L^AT_EX***. Look in the template to see how you can get an unnumbered and numbered list.

Which name is used in ***L^AT_EX*** to get a numbered list?

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☒ enumerate ✓

☐ document

Submit

You have used 1 of 3 attempts

i Answers are displayed within the problem

Text in Latex

Now that you know how to structure your document, it is time to add text, figures and tables.

Text is the easiest to add in ***L^AT_EX***: you just type it at the place you want it to appear. You do not have to think about the layout or the font, as ***L^AT_EX*** will take care of this for you. This way, all your text will appear in the same style.

In the first line of the preamble, the default font size for your document is defined. In our template it is set to 11pt. So if you want a little bigger font throughout your report, you should replace it with, for example, 12pt.

If you want to use a larger of smaller font size *locally*, you can use the commands from the following table:

Command	Output
<code>{\tiny sample text}</code>	sample text
<code>{\scriptsize sample text}</code>	sample text
<code>{\footnotesize sample text}</code>	sample text
<code>{\small sample text}</code>	sample text
<code>\normalsize sample text</code>	sample text
<code>{\large sample text}</code>	sample text
<code>{\Large sample text}</code>	sample text

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<code>{\huge sample text}</code>	sample text
<code>{\Huge sample text}</code>	sample text

These commands and the text you want to have in the smaller/bigger size must be between curly brackets.

Besides changing the font size, you can also emphasise text, for which you can see the commands in the next table:

Command	Output
<code>\textbf{sample text}</code>	sample text
<code>\textit{sample text}</code>	<i>sample text</i>
<code>\underline{sample text}</code>	<u>sample text</u>
<code>\emph{sample text}</code>	<i>sample text</i>

These commands can be used together, so `\textbf{\textit{\underline{sample text}}}` would create ***sample text***.

You can also include coloured text with the use of the `\color{...}` command. This command works the same as font size commands. `{\color{red} Red text}` results in Red text.

Figures and tables

Figures and table require in **L^AT_EX** some effort to create them. Let us start with an example of the code for a figure:

<code>\begin{figure}[htbp]</code>
<code>\centering</code>
<code>\includegraphics[width=0.5\textwidth]{Rainbowfish1000x774.jpg}</code>
<code>\caption{{\bf Caption of the figure.} And more information on what is in the figure, e.g. about the parameters used.}</code>
<code>\label{figure: Photo Rainbowfish}</code>
<code>\end{figure}</code>

As you can see, you need (at least) six lines to include a figure. The figure is what **L^AT_EX**

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caption. You can use the command `\ref{figure: Photo Rainbowfish}` to refer back to the figure, just like with chapters and sections.

Tables in **L^AT_EX** are another example of floats. A basic code for a table you can find in the template. The positioning of tables is similar to that of figures.

Bibliography

As you are writing a scientific report, you will probably use references in your text. **L^AT_EX** has the command `\cite{...}` to put these in the text. You must also add to what you want to refer for **L^AT_EX** to understand the references. This you do by adding a bibliography to the end of your document:

```
\begin{thebibliography}{10} % This number indicates the number of digits in
the maximum number of items in the bibliography
\addcontentsline{toc}{chapter}{Bibliography} %Adds a line to table of
contents
\bibitem{bib: name} All information used, that was not your own must have a
reference, so that the reader can recover the underlying information. When
sources are used or mentioned in the text, a reference must be made. % Refer
to this item in the text with \cite{bib: name}.
\bibitem{bib: another name} Leslie Lamport,
\textit{\LaTeX: a document preparation system},
Addison Wesley, Massachusetts,
2nd edition,
1994.
\end{thebibliography}
```

This is the most basic way to use references in **L^AT_EX**. A more advanced tool you can use is **B_BT_EX**, on which you can a good introduction here: **B_BT_EX**

Appendices

After you have added all relevant content to your report, you can also include appendices. In **L^AT_EX** appendices are similar to chapters, but are numbered differently. To start adding appendices, you include the command `\appendix` just before the first appendix you want to make. After this command, all chapters will be appendices, which you can start with the same command you used for making chapters.

If you want more information about **L^AT_EX**, we recommend you start at <https://en.wikibooks.org/wiki/LaTeX>, where you can find a lot of information

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Many people are using ***L^AT_EX***, so if you are stuck, ask your question to your favourite search engine.

Now you know the basics of ***L^AT_EX***, it is time to get to know one of the most powerful aspects of ***L^AT_EX***: typesetting mathematics. Go to the next page.

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